LESSON 01 / 03

Status of this sheet

Part I derived total, reflexive, symmetric and reflexive+symmetric counts with direct pair analysis. This sheet completes the catalogue (antisymmetric, equivalence/Bell, surjections by inclusion–exclusion) and shows the small-case listings an IOQM/RMO solution must carry. All counts are **models built from the Part I supply list** (product rule, complements, inclusion–exclusion on sets) — the technique is exam-derivable even where the topic is listed as an extension.

The catalogue for $|A| = n$

KEY RESULT

On an n-element set: relations $= 2^{n^2}$; reflexive $= 2^{n^2-n}$; symmetric $= 2^{n(n+1)/2}$; antisymmetric $= 2^n \cdot 3^{n(n-1)/2}$; reflexive+symmetric $= 2^{n(n-1)/2}$; reflexive+transitive = number of "preorder with fixed diagonal" (no simple closed form — casework); equivalence relations $= B_n$ (Bell): $1, 2, 5, 15, 52$ for $n = 1..5$. Functions $A \to B$ with $|B| = m$: m^n; injective: $\frac{m!}{(m-n)!}$ if $n \le m$ else 0; surjective: $n!\,S(n,m)$ with S the Stirling number $= \sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$.

SHORT DERIVATION

Surjections by inclusion–exclusion

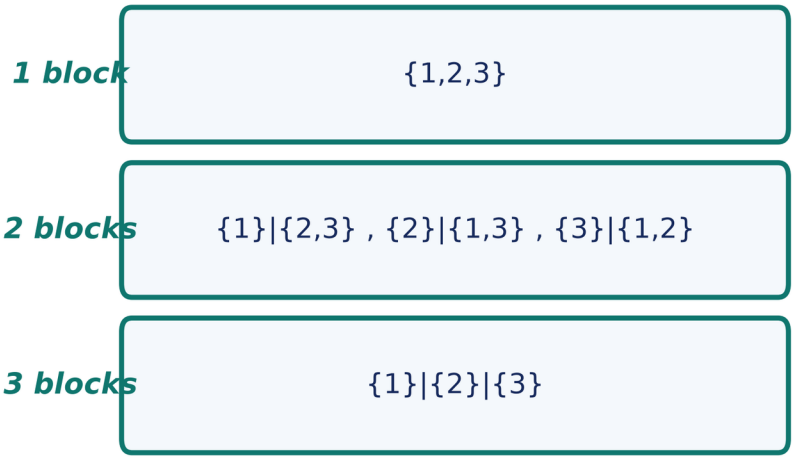
$f : A \to B$ surjective iff no element of B is missed. Let U = all m^n functions and E_b = functions missing the value b. We want $|U \setminus \bigcup_b E_b|$. Now $|E_b| = (m-1)^n$ (values chosen from the rest), $|E_b \cap E_c| = (m-2)^n$, and so on, so inclusion–exclusion gives $\sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$. Small case $n = 3, m = 2$: $2^3 - 2 \cdot 1^3 = 6$ ✓ matches the 6 onto functions $\{1,2,3\} \to \{a,b\}$ found by cases in Part I. Sanity: $m > n$ gives 0 (e.g. $n = 2, m = 3$: $9 - 3 \cdot 4 + 3 \cdot 1 = 0$).

WORKED EXAMPLE

Worked. Onto functions $\{1,2,3,4\} \to \{a,b,c\}$: $3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 - \binom{3}{3}0^4 = 81 - 48 + 3 = 36$. Cross-check with Stirling: $S(4,3) = 6$ (partitions of 4 elements into 3 blocks: one block of 2 and two singletons, $\binom{4}{2} = 6$), then order the blocks onto a, b, c: $6 \cdot 3! = 36$ ✓. Two independent methods agreeing is the standard olympiad "check" line.

Bell numbers by listings (do this in solutions)

For $A = \{1, 2, 3, 4\}$, $B_4 = 15$ partitions: 1 block (1 way), 2 blocks: choose the block containing 1 ($S(4, 2) = 7$), 3 blocks ($S(4, 3) = 6$), 4 singletons (1). Total $1 + 7 + 6 + 1 = 15$ ✓. General recurrence (extension): $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$ — element $n + 1$ joins a block after k others are already placed... more cleanly: $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}$: choose which elements share the block of $n + 1$.



The 5 equivalence relations on {1,2,3} as partitions (Bell B_3 = 5)

COMMON TRAP

- **Labelled vs unlabelled.** $\{\{1, 2\}, \{3, 4\}\}$ and $\{\{1, 3\}, \{2, 4\}\}$ are different partitions of a labelled set; counting unlabeled block shapes is a different (smaller) problem. State which one you count.
- Antisymmetric count $2^n 3^{n(n-1)/2}$: each off-diagonal unordered pair $\{a, b\}$ allows exactly 3 states — neither direction, only $a \to b$, only $b \to a$ — never 4 (both directions are banned).
- Inclusion–exclusion signs alternate. Boundary convention: a function on an empty domain is unique (the empty function), so $m^0 = 1$ for every m, including $0^0 = 1$ — state the convention whenever $n = 0$ appears.
- Surjections need $n \geq m$; injections need $n \leq m$. A formula that spits out a nonzero answer violating these is being used wrong.

ASSUMPTIONS & CONVENTIONS

Counts assume **finite** sets and labelled elements throughout. $S(n, m)$ here means Stirling numbers of the second kind (onto functions divided by $m!$); the notation is Part II vocabulary, but every *numerical* claim above is checkable by listing for $n, m \le 4$ — an olympiad write-up should always include one such listing.

LESSON 02 / 03

Status of this sheet

The equations below are **additional theory** (classic RMO/INMO and AIME-adjacent material). Nothing here contradicts the syllabus: every model shown is solvable with the Part I supply list plus one extra idea, stated openly. Do not claim such an equation "can never appear in JEE" — exam papers have historically leaked olympiad-flavoured models; what is not promised is any theorem beyond the statements given here.

The Cauchy equation $f(x+y)=f(x)+f(y)$ on $\mathbb{R}$

SHORT DERIVATION

Rational linearity

$f(0)=f(0+0)=f(0)+f(0)\Rightarrow f(0)=0$. Induction: $f(nx)=nf(x)$ for $n\in\mathbb{N}$; then $0=f(0)=f(x+(-x))=f(x)+f(-x)\Rightarrow f(-x)=-f(x)$, so $f(nx)=nf(x)$ for all integers n. For p/q with $q\geq 1$: $f(p)=f\left(q\cdot\frac{p}{q}\right)=qf\left(\frac{p}{q}\right)\Rightarrow f\left(\frac{p}{q}\right)=\frac{p}{q}f(1)$. Hence $f(r)=cr$ for every rational r, where $c=f(1)$.

KEY RESULT

Regular upgrade. If f is additionally monotone on any interval, or continuous at a point, or bounded above/below on any interval, then $f(x)=cx$ for all real x (rational-linearity plus density of $\mathbb{Q}$ and the squeeze). Without some regularity, non-linear solutions exist (via a Hamel basis; they are non-measurable and cannot be written down explicitly) — so: **never write $f(x)=cx$ as "the" solution unless the problem grants a regularity condition.** Many exam models quietly assume continuity; state it.

WORKED EXAMPLE

Worked models.

1. $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{Q}$, $f(1)=2$: then $f(7/3)=\frac{7}{3}\cdot 2=\frac{14}{3}$ — no regularity needed, because the domain is $\mathbb{Q}$ itself (rational linearity already complete).
2. $f:\mathbb{R}\to\mathbb{R}$ continuous, $f(x+y)=f(x)+f(y)$, $f(2)=6$: $c=f(1)=3$, so $f(x)=3x$ for every real x. **Check:** $f(x+y)=3(x+y)=3x+3y$ ✓.
3. $f:\mathbb{R}\to\mathbb{R}$ with $f(x+y)=f(x)+f(y)$ and $f(x)\geq 0$ for all $x>0$ (bounded below on $(0,\infty)$): the regularity condition is what upgrades $f(r)=cr$ to $f(x)=cx$; then $f(x)\geq 0$ for $x>0$ forces $c\geq 0$.

The multiplicative cousins

KEY RESULT

$f(x+y)=f(x)f(y)$: either $f\equiv 0$, or $f(0)=1$ and f never vanishes; with continuity $f(x)=a^x$, $a=f(1)>0$. $f(xy)=f(x)+f(y)$ on $(0,\infty)$: with continuity $f(x)=c\ln x$ (set $g(t)=f(e^t)$ and Cauchy reduces). $f(xy)=f(x)f(y)$ on $(0,\infty)$: $f(x)=x^c$ or $f\equiv 0,1$ specials. **Pexider:** $f(x+y)=g(x)+h(y)$ for all x,y implies all three are affine with $f=g+h+\text{const}$ relationships — plug $y=0$ to express g,h through one function and fall back to Cauchy. **Jensen:** $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}$ with continuity $\Rightarrow f(x)=cx+d$.

SHORT DERIVATION

$f(x+y)=f(x)f(y)$ quickly

Put $y=0$: $f(x)=f(x)f(0)$, so either $f\equiv 0$ or $f(0)=1$. If $f(x_0)=0$ for some x_0, then $f(x)=f((x-x_0)+x_0)=f(x-x_0)f(x_0)=0$ for all x: $f\equiv 0$. So a non-zero solution never vanishes; if it takes both signs it still can't cross 0 (continuous image of connected set), hence a positive solution has $f(x)=e^{g(x)}$ with g additive, and continuity finishes $g(x)=x\ln a$.

Involutions $f(f(x))=x$

$f\circ f=\mathrm{id}$ means f is its own inverse: f is bijective and $f^{-1}=f$. Solutions of $f(f(x))=x$ split into **fixed points** ($f(x)=x$) and **2-cycles** ($f(x)=y\neq x$, $f(y)=x$).

SHORT DERIVATION

Increasing involution fixes everything

Suppose f is strictly increasing and $f(f(x))=x$. If $f(x_0)>x_0$ then applying the increasing f gives $f(f(x_0))>f(x_0)>x_0$, i.e. $x_0>f(x_0)$ — contradiction. Similarly $f(x_0)<x_0$ contradicts. So $f(x)=x$ for all x: every increasing involution is the identity on its domain. A decreasing involution (e.g. $f(x)=a-x$, $f(x)=1/x$ on $\mathbb{R}^+$) instead swaps pairs around fixed points.

WORKED EXAMPLE

Worked. Find all continuous $f:\mathbb{R}\to\mathbb{R}$ with $f(f(x))=x$ and f decreasing. Any x with $f(x)=x$ is a fixed point; $g(x)=f(x)-x$ is decreasing (f dec, $-x$ dec... sum of two decreasing is decreasing) hence g has at most one zero — at most one fixed point. Example $f(x)=a-x$ has exactly one ($x=a/2$). The general decreasing-continuous involution picture: one fixed point (if g changes sign) and the rest of the line paired off in 2-cycles symmetric about it. **Check:** $f(x)=1/x$ on $(0,\infty)$: decreasing, $f(f(x))=x$ ✓, fixed point $x=1$ ✓.

COMMON TRAP

- Cauchy with domain $\mathbb{Q}$: rational linearity is the complete answer. Cauchy with domain $\mathbb{R}$: rational linearity + "regularity" is required; state the condition.
- $f(x) = x^2$ is even, $f(x) = x^3$ odd, but $f(x) = x^2 + x^3$ is neither — decomposition (Part I) is the safe tool.
- In $f(f(x)) = x$ problems, plugging $x = f(t)$ is the standard trick: $f(f(f(t))) = f(t)$, i.e. $f^3 = f$; combined with $f^2 = \mathrm{id}$ this just re-proves bijectivity — write the injection/surjection argument once, cleanly.
- Periodic + additive: $f(x + T) = f(x)$ with additivity forces $f(T) = f(0) = 0$, and with continuity that collapses $f \equiv 0$ (since $f(r) = cr$ and $cT = 0$). Without continuity a wild additive T-periodic solution can still exist on a Hamel basis — so again: state the regularity condition before claiming collapse. The point stands: ***imposing two strong structures usually collapses the family to trivial members, but "usually" needs a hypothesis.***

LESSON 03 / 03

Status of this sheet

The greatest-integer function is **in the official syllabus** (listed under special functions); Hermite's identity and the fixed-point decomposition are extra theory. Every worked model below is solved with syllabus tools only — so treat the identities as accelerators, and the models as fair game.

Hermite's identity

SHORT DERIVATION

Sum of n shifted floors

For real x and positive integer n:

$$\sum_{k=0}^{n-1} \lfloor x + \tfrac{k}{n} \rfloor = \lfloor nx \rfloor$$

Write $x = m + \theta$ with $m = \lfloor x \rfloor$, $\theta \in [0, 1)$. Then $\lfloor x + \frac{k}{n} \rfloor = m + \lfloor \theta + \frac{k}{n} \rfloor$, so the left side is $nm + \sum_k \lfloor \theta + \frac{k}{n} \rfloor$. Now $\lfloor \theta + \frac{k}{n} \rfloor = 1$ exactly when $\frac{k}{n} \geq 1 - \theta$, i.e. $k \geq n(1-\theta)$; the number of $k \in \{0, \ldots, n-1\}$ with $k \geq n(1-\theta)$ is $\lfloor n\theta \rfloor$ (boundary $n(1-\theta)$ an integer counts k from that value up). So the sum $= nm + \lfloor n\theta \rfloor = \lfloor nm + n\theta \rfloor = \lfloor nx \rfloor$ ✓.

WORKED EXAMPLE

Worked checks. $n = 3$, $x = 1.7$:
$\lfloor 1.7 \rfloor + \lfloor 1.7 + 0.33 \rfloor + \lfloor 1.7 + 0.67 \rfloor = 1 + 2 + 2 = 5 = \lfloor 5.1 \rfloor$ ✓. $x = 2$ exactly:
$2 + 2 + 2 = 6 = \lfloor 6 \rfloor$ ✓. **Check boundary:** when $n\theta$ is an integer the threshold $k \geq n(1-\theta)$ is attained (included) — $\lfloor \cdot \rfloor$ inequality is $\geq$ on both sides of the threshold count.

Floor identities and equations

KEY RESULT

$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$; $\lfloor x \rfloor + \lfloor y \rfloor \leq \lfloor x + y \rfloor \leq \lfloor x \rfloor + \lfloor y \rfloor + 1$; $\lfloor x \rfloor + \lfloor -x \rfloor = 0$ if $x \in \mathbb{Z}$ else -1; $\{x\} + \{-x\} = 0$ or 1 with the same split; number of integers in $(a, b]$ is $\lfloor b \rfloor - \lfloor a \rfloor$; $\lfloor x \rfloor = c \Leftrightarrow x \in [c, c+1)$; $\lfloor x \rfloor < c \Leftrightarrow x < c$ (for integer c); $\lfloor x \rfloor \geq c \Leftrightarrow x \geq c$ (integer c).

STANDARD APPLICATION

Model (fully syllabus-level). Solve $\lfloor x^2 \rfloor = 2$ over the reals. Sandwich: $2 \le x^2 < 3$. Left half: $x^2 \ge 2 \Rightarrow |x| \ge \sqrt{2}$; right half: $x^2 < 3 \Rightarrow |x| < \sqrt{3}$. Solution set $= \{x : \sqrt{2} \le |x| < \sqrt{3}\} = [-\sqrt{3}, -\sqrt{2}] \cup [\sqrt{2}, \sqrt{3})$ — **closed** at $\pm\sqrt{2}$ ($x^2 = 2$ gives $\lfloor 2 \rfloor = 2$ ✓) and **open** at $\pm\sqrt{3}$ ($x^2 = 3$ would give $\lfloor 3 \rfloor = 3$ ✗). The answer is a set of reals, not a number. Integer version: $x \in \mathbb{Z}$ needs $x^2 \in [2, 3)$, but squares of integers are $0, 1, 4, \ldots$ — **no solutions**. **Check:** the two phrasings ("all real x" vs "all integer x") have completely different answers; the quantifier is part of the question.

Fixed points and involutions (counting flavour)

$f(x) = x$ is a **fixed point**; $f(f(x)) = x$ is an **involution equation** whose solution set decomposes into fixed points of f plus 2-cycles $\{x, f(x)\}$ with $f(x) \neq x$.

KEY RESULT

If f is strictly increasing and $f(f(x)) = x$ on an interval, then $f(x) = x$ everywhere on it (Part II sheet 02 proof). If f is strictly decreasing and continuous, $g(x) = f(x) - x$ is strictly decreasing, so f has **at most one** fixed point; each non-fixed x pairs with $f(x)$ in a 2-cycle. Counting solutions of $f(f(x)) = x$ for a concrete monotone f therefore reduces to: find fixed points (solve $f(x) = x$), then account for the symmetric 2-cycles.

WORKED EXAMPLE

Worked. $f(x) = 3 - x$ on $\mathbb{R}$: decreasing, $f(f(x)) = 3 - (3 - x) = x$ ✓ for **every** x — the entire line solves the involution equation, yet only $x = 3/2$ is a fixed point. Moral: "solutions of $f(f(x)) = x$" and "fixed points of f" are different sets; the question wording decides which is asked. With $f(x) = \frac{1}{x}$ on $(0, \infty)$: $f(f(x)) = x$ everywhere; fixed point $x = 1$ only. **Check:** $f(2) = 1/2$, $f(1/2) = 2$ — a 2-cycle, as predicted by the decreasing-case analysis.

COMMON TRAP

- $\lfloor x \rfloor$ of a negative number: $\lfloor -1.2 \rfloor = -2$, not -1 (greatest integer **less or equal**). Truncation toward zero is a different function.
- In floor equations, the answer set's endpoints are decided by strict vs non-strict inequalities — write the sandwich first, then convert to interval notation, then test both endpoints.
- $\lfloor x^2 \rfloor = 2$ vs $\lfloor x \rfloor^2 = 2$: the second means $\lfloor x \rfloor = \pm\sqrt{2}$, impossible for integer $\lfloor x \rfloor$ — no solutions. Composition order matters even inside floor equations.
- Never square an inequality $\sqrt{2} \le |x| < \sqrt{3}$ without tracking that $|x|$ is already non-negative — squaring is safe on $[0, \infty)$ but flips nothing here; for $x < 0$ reflect by $|x|$.

ASSUMPTIONS & CONVENTIONS

All floor statements are for real x; n is a positive integer in Hermite; fixed-point claims assume the stated monotonicity/continuity on an interval. 2-cycles are unordered pairs $\{x, y\}$ with $y = f(x) \neq x$; when "count solutions" is asked on a finite set, count elements x, not cycles.